



Develop and Development of an AI based Chatbot ERP Navigation System

A high-performance, AI-assisted School Management Platform



Project Overview

What is ERP System?

ERP (Enterprise Resource Planning) in a school is a centralized, digital software system that integrates all academic and administrative processes into one platform. It manages daily operations like admissions, student attendance, fee collection, grades, and parent-teacher communication in real-time, reducing paperwork and enhancing data security.



Live Dashboard

Real-time institutional stats & KPIs



Student Management

Full CRUD with filtering & search



Analytics & Reports

Attendance, fees & revenue tracking

Key Features



Admin Dashboard

Institutional overview with live stats: total students, active teachers, daily attendance rate, and monthly revenue — all at a glance.



Student Directory

Complete student management with CRUD operations, grade filtering, status badges, attendance bars, and fee payment tracking.



Smart Search & Filter

Global search bar for students and teachers. Multi-criteria filtering by grade, status, and payment — with instant results.



Notifications & Alerts

Unread notification indicators for principals. Real-time feedback with budget warnings and activity alerts.



Academic Calendar

Scheduled events, board meetings, and sports meets displayed in the active schedule panel with day and time details.



New Student Registration

Animated modal form for enrolling new students with grade selection and instant UI update on submission.

What kind of assets and resources a school ERP system should manage?

Usually, school building, furniture, library books, laboratory equipment, office and classroom stationery, medical supplies, physical training instruments, sports gear, school staff and student uniforms, computers and electronics, cleaning materials, vehicles are just a few examples of the wide range of inventory that an educational institute may hold. All those assets have to be entered into the system; their purchase dates, expiry, maintenance reports have to be captured for assets deployment and management.

Dashboard for the School Management

This feature allows the school management to list school courses and upload course materials. Integration with the student-parent and staff or faculty management systems enables the management to map courses to the relevant classes and assign appropriate staff to the course. Calendar management system integration allows the power to create and schedule a timetable and give the course materials. Integration with the accounting management system should enable the management to overview the financial data about a specific course or courses.

Students can log in to their accounts and see an exclusive preview of their school activities. They can download or view the class notes and lecture audio or video, upload their assignments, etc. They can view their class schedule for the current period and plan their studies.

Dashboard Analytics



4+

Students
Enrolled



2

Active
Teachers



94.2%

Daily
Attendance

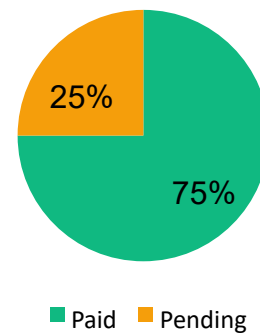
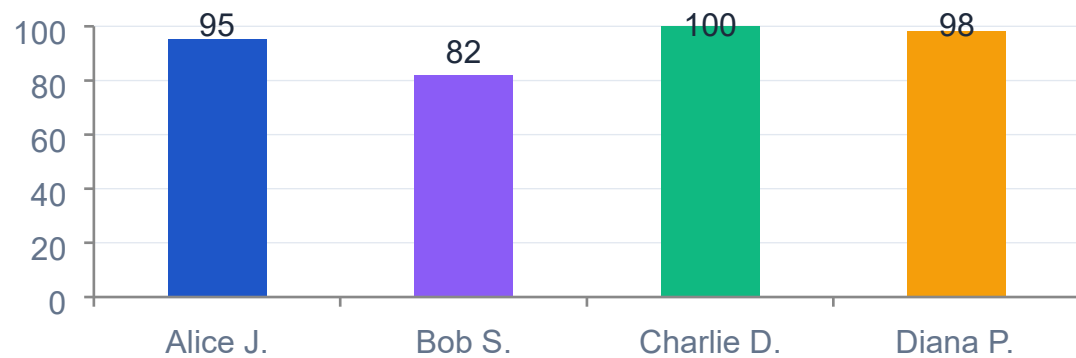


\$45.2K

Monthly
Revenue

Student Attendance Overview

Fee Payment Status



WHY ?

The system offers high performance for all departments in the school to constitute relationship among them and drive superior results. The authorized users can have access to the integrated database from anywhere and at any time. The system promotes efficiency, user satisfaction and collaboration of different departments.

Challenges & Solutions

Challenge

State became inconsistent after adding a student — UI didn't update automatically.

Sidebar animation caused layout shift and jitter on toggling.

Solution

Switched to React `useState` with immutable spread patterns; new student is unshifted into state array for instant re-render.

Used Framer Motion's `AnimatePresence` with x-axis enter/exit animation and CSS transition on main content padding.

TypeScript type errors between frontend expectations and server response shape.

Vite dev server conflicted with Express API routes causing 404s.

Centralized shared types in `types.ts` used by both API handlers and React components — single source of truth.

Configured Vite in `middlewareMode: true` inside Express, so Express handles `/api/*` first and Vite handles the rest.

Future Scope & Roadmap

Phase 1

Database Integration

- ✓ MongoDB/PostgreSQL persistence
- ✓ User auth with JWT tokens
- ✓ Password hashing with bcrypt

Phase 2

Academic Module

- ✓ Grade book & result management
- ✓ Subject & curriculum tracking
- ✓ Exam scheduling system

Phase 3

Communication Hub

- ✓ Email/SMS notifications
- ✓ Parent portal access
- ✓ Real-time chat with teachers

Phase 4

AI & Insights

- ✓ Gemini AI chatbot integration
- ✓ Dropout risk prediction
- ✓ Personalized learning paths



Thank You

Built with passion for modern education management